

A day in the field

One drillhole, from setting up the project to exporting the data. Follow it step by step on your phone. Every value to type is here, and so is what the app should show when you get it right.

PROJECT

**Masbate Gold
Pilot**

HOLE

MB-DDH-001

FINAL DEPTH

45.3 m

TIME

**about one
working day**

What this is

A realistic day for a field geologist working alone: the drillers have finished a 45.3 m core hole, and you log it and sample it. It is fictional data for testing, so nothing here is a real result and no real coordinates are used.

It is also our official demo of the app. If the app can do this whole day smoothly, with no signal, it is ready for a real hole.

What you need

- The CoreChain Field app on an Android phone, charged
- Camera and location allowed when the app asks
- Something to photograph as "core boxes": a table, a shelf, real boxes if you have them
- A computer with Excel or Google Sheets for Part 8
- This guide printed, or open on a second screen

How to use it

- Do the parts in order. Tick each step as you go.
- Turn on **airplane mode** before you start. The app must work with no signal.
- Green **You should see** boxes are checkpoints. If the app shows something different, take a screenshot and write the step number on the sign-off sheet.
- Anything that confused you counts, even if it "worked".

The day at a glance

Target times are what a practised user should manage. Note your own on the sign-off sheet.

Part	What you do	What you record	Target time
1	Set up the project and the hole	1 project, 1 hole	10 min
2	Core reception: enter the drillers' runs and the core boxes	15 runs, 9 boxes	45 min
3	Photograph the core	11 photos	25 min
4	Log the core	12 intervals	60 min
5	Sample the core, with QC	25 samples (22 primary + 3 QC)	75 min
6	Close out the hole	Status "Logged"	5 min
7	Trace samples back to the core	4 samples checked	10 min
8	Export the data and check it in Excel	5 CSV files	15 min
9	Try to break it	A scratch hole, 16 steps	20 min
Total			about 4 h 25 min

At the end of the day you should have

15 core runs	94.5% core recovery	9 core boxes	11 photos
12 logged intervals	45.3 of 45.3 m core logged	25 samples	1 / 1 / 1 standard / blank / duplicate

The story of the hole, in one look

Depth (m)	What the core is	Why it matters for the test
0–3	Soft, clay-rich saprolite	Poor recovery (60%). Tests low recovery and box 1 notes.
3–14	Altered andesite	Weathering changes from highly to moderately weathered.
18–21	Fault breccia	Recovery 90%, RQD 53%. Photographed close up. Structure code FLT.
21–33	Silicified diorite with a quartz stockwork	The mineralised zone: sampled every metre.
27–28.2	Quartz vein with visible gold	The sample that gets a duplicate. Traced back in Part 7.
33–45.3	Fresh granodiorite	Sampled to 40 m, then no more. End of hole at 45.3 m.

1 Set up the project and the hole

About 10 minutes. Airplane mode on.

- ☐ **1.1** Open CoreChain. On the home screen tap **New project** and fill in:

Field	Enter
Project name	Masbate Gold Pilot
Coordinate system	Tap the WGS84 chip
Commodity	Gold
Location	Masbate, Philippines
Sample ID prefix	MB

Tap **Create project**.

- ☐ **1.2** Open the project and tap **Settings**. Look, but do not change anything.

YOU SHOULD SEE

- Sample ID prefix
MB
, Next sample number
1
- Standard, Blank and Field duplicate: every
20
samples
- Largest photo size
1.5 MB

- ☐ **1.3** Back on the project screen, tap **New drillhole** and fill in:

Field	Enter
Hole ID	MB-DDH-001
Planned depth (m)	45
Planned azimuth (0-360°)	90
Planned inclination / dip (-90 to 90°)	-60
Collar	Outdoors: tap Use GPS and wait for the fix. Indoors: type Latitude 12.3714 and Longitude 123.5987

Tap **Create drillhole**, then open it from the list.

- ☐ **1.4** The drillers finished yesterday, so you already know the final depth. On the hole screen:
- Tap the **Complete** status chip.
 - In **Actual details**: tap **Started** and pick the date **3 days ago** in the calendar. Tap **Completed** and pick **yesterday**.
 - **Actual final depth (m)**: **45.3**. Tap **Save actual details**.

YOU SHOULD SEE

- Title
MB-DDH-001
with a
COMPLETE
pill
- "Masbate Gold Pilot · 45.3 m final · azimuth 90° · dip -60°"
- A collar line (GPS or the coordinates you typed)
- Core logged
0 OF 45.3 M
, and the big button says
ADD FIRST CORE BOX

2 Core reception

About 45 minutes. The rig has sent up the core in runs of 3 m; you enter what the driller wrote on each block, then the boxes.

- ☐ **2.1** On the hole screen tap the **Core runs** tile, then **Add run**. Enter the runs in order. The from-depth fills in for you after the first one. Under the depths, the **3 m** chip sets the to-depth for you, and **Full recovery** fills in the recovered length.

Run	From (m)	To (m)	Recovered length (m)	Pieces $\geq$ 10 cm, total length (m)	Recovery	RQD
1	0.0	3.0	1.80	0.40	60%	13.3%
2	3.0	6.0	2.40	1.20	80%	40%
3	6.0	9.0	2.85	2.00	95%	66.7%
4	9.0	12.0	3.00	2.40	100%	80%
5	12.0	15.0	2.94	2.10	98%	70%
6	15.0	18.0	3.00	2.60	100%	86.7%
7	18.0	21.0	2.70	1.60	90%	53.3%
8	21.0	24.0	2.91	2.20	97%	73.3%
9	24.0	27.0	3.00	2.70	100%	90%
10	27.0	30.0	3.00	2.80	100%	93.3%
11	30.0	33.0	2.97	2.50	99%	83.3%
12	33.0	36.0	3.00	2.90	100%	96.7%
13	36.0	39.0	3.00	2.85	100%	95%
14	39.0	42.0	2.94	2.60	98%	86.7%
15	42.0	45.3	3.30	3.00	100%	90.9%

The last two columns are worked out by the app. **Do not type them.** Use them to check your entries.

YOU SHOULD SEE

- Run 1 shows recovery
60%
, RQD
13.3%
- Run 7 shows
90%
and
53.3%
- No amber gap or overlap warnings while entering any run.
- On the hole screen the Core runs tile says
15 RUNS · 94.5% RECOVERY

- ☐ **2.2** Back on the hole screen tap **Core boxes**, then **Add box**. The box number and from-depth fill in for you, and the **5 m** chip sets the to-depth.

Box number	From (m)	To (m)	Note (optional)
1	0.0	5.0	Clay-rich, broken, wet
2	5.0	10.0	–
3	10.0	15.0	–
4	15.0	20.0	Fault zone 19.2 to 19.5 m, gouge
5	20.0	25.0	–
6	25.0	30.0	Quartz vein 27.0 to 28.2 m, visible gold
7	30.0	35.0	–
8	35.0	40.0	–
9	40.0	45.3	End of hole

YOU SHOULD SEE

- The Core boxes tile says
9 BOXES
.
- No "Gap in core boxes" warning on the hole screen.
- The last box ends at
45.3 M
, the same depth as the last run.

TIP

If you make a mistake in a run or a box, delete it and add it again. Editing an existing entry is not built yet.

3 Photograph the core

About 25 minutes. The first time you take a photo, the phone asks for camera permission: choose **While using the app**.

Each photo is filed automatically against the box or interval you took it from, with the hole ID, box number, depth range and time. You do not name anything. Photos are shrunk to under 1.5 MB on the phone.

Done	Photo of	Depth	Where to tap	What to include
☐	Box 1	0–5 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 2	5–10 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 3	10–15 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 4	15–20 m	Core boxes → Photos (0) under the box → Take photo	Include the gouge at 19.2 to 19.5 m
☐	Box 5	20–25 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 6	25–30 m	Core boxes → Photos (0) under the box → Take photo	Include the quartz vein at 27.0 to 28.2 m
☐	Box 7	30–35 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 8	35–40 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Box 9	40–45.3 m	Core boxes → Photos (0) under the box → Take photo	Lid open, core wet, label in view
☐	Interval	18–21 m	Core log → Photos (0) on the interval → Take photo (after Part 4)	Close-up of the fault breccia
☐	Interval	27–28.2 m	Core log → Photos (0) on the interval → Take photo (after Part 4)	Close-up of the vein and the gold

YOU SHOULD SEE

- The camera screen shows the label at the top, for example
MB-DDH-001 · BOX 6 · 25–30 M
- After the shutter, you go back to the photo list with a thumbnail, the time, the size (well under 1.5 MB) and **DELETE**
- Every box now says **PHOTOS (1)**

TIP

Hold the phone flat above the box. Photograph the whole box, not just one row. If you get a bad photo, tap Delete and take it again.

4 Log the core

About 60 minutes. Core log → **Add interval**. The from-depth fills in with the end of the last interval. Tap a code chip, or type the code. The **1 m, 2 m, 3 m, 5 m** chips set the to-depth.

#	From	To	Lithology	Alteration type	Alteration intensity	Mineralisation: mineral	Mineralisation: style	Mineral content (%)	Weathering	Structure type	Structure notes and comments
1	0	3	OVb	–	–	–	–	–	CW	–	Saprolite, clay-rich, orange-brown. Core broken.
2	3	8	AND	ARG	3	–	–	–	HW	–	Limonite on fractures. Original texture still visible.
3	8	14	AND	PHY	2	PY	DISS	2	MW	–	–
4	14	18	AND	PHY	3	PY	VNL	3	SW	–	Sericite-pyrite veinlets, 1 to 3 mm.
5	18	21	BX	PHY	3	PY	BXH	4	SW	FLT	Fault breccia. Gouge at 19.2 to 19.5 m. Poor recovery.
6	21	27	DIO	SIL	4	PY	STK	5	FR	–	Quartz stockwork, about 8 veinlets per metre.
7	27	28.2	QV	SIL	4	AU	VNL	0.1	FR	VEIN	Vuggy quartz vein. Visible gold at 27.6 m.
8	28.2	33	DIO	SIL	3	PY	STK	4	FR	–	–
9	33	36	DIO	PHY	2	PY	DISS	2	FR	–	–
10	36	39.5	GRD	PROP	2	–	–	–	FR	–	Chlorite and epidote after biotite.
11	39.5	40.5	GRD	PROP	2	–	–	–	FR	FRAC	Calcite-filled fractures.
12	40.5	45.3	GRD	PROP	1	–	–	–	FR	–	Fresh granodiorite to end of hole.

TIP

Intervals **4** and **12** are almost the same as the one before. Tap **Copy previous interval**, then change only what differs: interval 4 (intensity 3, style VNL, 3%, weathering SW, new note); interval 12 (intensity 1, no structure, new note).

TIP

Take a call in the middle of an interval? Close the form. Your half-typed interval is saved, and comes back when you open **Add interval** again.

YOU SHOULD SEE

- After the last interval the hole screen says
CORE LOGGED 45.3 OF 45.3 M
and the progress bar is full.
- The depth strip is solid blue: no amber gaps, no red overlaps. The Core log tile says
12 INTERVALS
.
- No amber warning row on the hole screen at all.

Now go back to Part 3 and take the two interval photos (18–21 m and 27–28.2 m).

5 Sample the core

About 75 minutes. Samples → **New sample**. Choose the hole chip **MB-DDH-001** and the type chip. The **Sample number** is filled in for you and should match the table. For a primary sample the **From depth** is filled in with where your last sample ended, so you only set the to-depth (type it, or tap the **1 m** or **2 m** chip). Check that it matches the table.

#	Sample number	Type	Depth (m)	Also enter	Note (optional)	Done
A. Primary samples, in order (samples 1 to 20)						
1	MB-00001	Primary	14–16		–	☐
2	MB-00002	Primary	16–18		–	☐
3	MB-00003	Primary	18–19		–	☐
4	MB-00004	Primary	19–20		–	☐
5	MB-00005	Primary	20–21		–	☐
6	MB-00006	Primary	21–22		–	☐
7	MB-00007	Primary	22–23		–	☐
8	MB-00008	Primary	23–24		–	☐
9	MB-00009	Primary	24–25		–	☐
10	MB-00010	Primary	25–26		–	☐
11	MB-00011	Primary	26–27		–	☐
12	MB-00012	Primary	27–28.2		Quartz vein, visible gold at 27.6 m	☐
13	MB-00013	Primary	28.2–29.2		–	☐
14	MB-00014	Primary	29.2–30.2		–	☐
15	MB-00015	Primary	30.2–31.2		–	☐
16	MB-00016	Primary	31.2–32.2		–	☐
17	MB-00017	Primary	32.2–33		–	☐
18	MB-00018	Primary	33–34		–	☐
19	MB-00019	Primary	34–35		–	☐
20	MB-00020	Primary	35–36		–	☐

#	Sample number	Type	Depth (m)	Also enter	Note (optional)	Done
B. Stop. Open the Samples register: three amber reminders appear. Insert the controls (samples 21 to 23)						
21	MB-00021	Standard	none	Reference material ID: OREAS 45e	Use the certified standard your lab supplies	☐
22	MB-00022	Blank	none		Barren silica sand	☐
23	MB-00023	Duplicate	copied	Duplicate of: MB-00012	Quarter core, same interval as MB-00012	☐
C. Back to the primaries (samples 24 and 25)						
24	MB-00024	Primary	36–38		–	☐
25	MB-00025	Primary	38–40		–	☐

YOU SHOULD SEE

- After sample 20 the register shows three amber cards:
A STANDARD IS DUE
,
A BLANK IS DUE
,
A FIELD DUPLICATE IS DUE
, each with "20 samples since the last one (target: every 20)". Before sample 20 none are shown.
- Tap **INSERT STANDARD**
(and the others) to jump to a new-sample form with the type already chosen. Once all three are saved the amber cards disappear.

YOU SHOULD SEE

- The register lists **25 SAMPLES**
. Filter **TYPE**
= Standard shows one.
- QC insertion rate:
STANDARDS: 1 (1 PER 22 SAMPLES, TARGET 1 PER 20)
, and the same for blanks and field duplicates.
- The hole screen Samples tile says **25 SAMPLES**
.

TIP

In real life the lab wants the standard hidden among the batch. This test only checks the reminder rule, so we insert them in one go after sample 20.

6 Close out the hole

About 5 minutes.

- ☐ **6.1** On the hole screen tap the **Logged** status chip.
- ☐ **6.2** Look at the whole hole screen, top to bottom.

YOU SHOULD SEE

- Pill:
LOGGED
. Core logged
45.3 OF 45.3 M
, no warnings.
- Tiles:
9 BOXES

15 RUNS · 94.5% RECOVERY

12 INTERVALS

25 SAMPLES

• Actual details: Started and Completed show the dates you picked, final depth
45.3
.

- ☐ **6.3** Go back to the home screen.

YOU SHOULD SEE

- **CONTINUE WHERE YOU LEFT OFF**
shows
MB-DDH-001
in
MASBATE GOLD PILOT
, 100% logged.
- Your project card shows
1 HOLE · GOLD · 100% LOGGED
.

- ☐ **6.4** Force-close the app (recent apps, swipe it away), then open it again. Everything is still there. This is the "battery died at the rig" test.

7 Trace samples back to the core

About 10 minutes. Samples → tap a sample. The screen shows where it has been, from the hole to the assay.

Open sample	Chain you should see
MB-00012 the quartz vein	Hole MB-DDH-001, GPS or typed collar Box 6 · 27–28.2 m , 1 photo Logged interval QV · SIL Sample created Bagged (next step) → Dispatched → Assay result

Open sample	Chain you should see
MB-00003 in the fault zone	Box 4 · 18–19 m , 1 photo Logged interval BX · PHY
MB-00025 crosses two intervals	Box 8 · 38–40 m , 1 photo Logged interval GRD · PROP (+1 more)
MB-00021 the standard	Starts at the hole (no core to trace) Sample created: Reference material OREAS 45e Bagged → Dispatched → Assay result

YOU SHOULD SEE

- Every green tick is real data. No amber "missing" warning on any sample in this hole.
- The

BAGGED

step is highlighted as next but cannot be tapped yet. Custody and dispatch come in a later update, so this is expected.

8 Export the data and check it

About 15 minutes. Project screen → **Export**. Tap **Share** on each file and save it to the phone (Files or Downloads). If you are still in airplane mode, that is enough for now; switch airplane mode off afterwards and send the files to yourself (email, Drive, USB). Open them on a computer.

File	Rows	Check in Excel
<code>masbate-gold-pilot-collars.csv</code>	1	HOLEID MB-DDH-001, LONGITUDE 123.5987, LATITUDE 12.3714, COORDINATE_SYSTEM WGS84, PLANNED_DEPTH 45, FINAL_DEPTH 45.3, STATUS logged
<code>masbate-gold-pilot-surveys.csv</code>	1	DEPTH 0, AZIMUTH 90, DIP -60 (planned values: no downhole surveys yet)
<code>masbate-gold-pilot-log.csv</code>	12	Depths run 0 to 45.3 with no gaps. Interval 7 has MINERAL_AU, MINERAL_PERCENT 0.1, STRUCTURE_TYPE VEIN.
<code>masbate-gold-pilot-runs.csv</code>	15	RECOVERY_PCT of run 1 is 60 and of run 7 is 90. Sum of RECOVERED_M is 42.81, sum of DRILLED_M is 45.3.
<code>masbate-gold-pilot-samples.csv</code>	25	SAMPLE_ID MB-00001 to MB-00025. QC column marks the three controls. MB-00023 has PARENT_SAMPLE_ID MB-00012 and the same FROM and TO.

YOU SHOULD SEE

- All five files open as clean tables: no odd characters, one header row, numbers as numbers.
- HOLEID is **MB-DDH-001** on every row.
- The Export screen shows the file names and row counts above **BEFORE** you share.

TIP

Core boxes and photos are not in the export yet. They are on the phone, and they show up in the sample trace.

9 Try to break it

About 20 minutes. Do these on the scratch hole **TEST-01** so the day's data stays clean. Each one checks that the app catches a mistake a tired person makes in the field.

	Do this	You should see	Where
☐	1. Create a second hole with ID MB-DDH-001 in this project.	"A hole named "MB-DDH-001" already exists in this project." No hole is created.	Project
☐	2. Create a scratch hole TEST-01 , planned depth 10 . Use it for everything below.	The hole is created. The rest of the day's data is untouched.	Project
☐	3. On TEST-01 add box 1 from 0 to 5, then box 2 from 6 to 10.	Amber "Leaves a gap of 5–6 m next to this box." The button now says Save anyway . Do not save it.	Boxes
☐	4. Add box 2 from 4 to 8.	Amber "Overlaps another box at 4–5 m." Do not save it.	Boxes
☐	5. Add a run from 0 to 3 with Recovered length 3.4 .	Amber "Recovered 3.4 m is more than the 3 m drilled. Check the entry before saving."	Runs
☐	6. Add a run from 0 to 3, recovered 2 , pieces ≥ 10 cm 2.5 .	Red "Pieces ≥ 10 cm can't be longer than the recovered core." You cannot save.	Runs
☐	7. Add an interval from 3 to 2.	Red "To depth must be greater than from depth."	Log
☐	8. Add an interval 0 to 2 with Mineral content 150 .	Red "Mineral % must be between 0 and 100."	Log
☐	9. Add a primary sample from 9 to 11.	Red "A sample must fall inside the hole's depth (0–10 m)."	Samples
☐	10. Save a primary sample 0 to 2, then try 1 to 3.	Red "Overlaps a primary sample at 0–2 m."	Samples
☐	11. Choose type Standard and leave Reference material ID empty.	Red "Enter the reference material ID."	Samples
☐	12. Choose type Duplicate and pick nothing under "Duplicate of".	Red "Choose the primary sample this duplicates."	Samples
☐	13. Type the sample number MB-00001 on TEST-01.	Red "MB-00001 is already used in this project." Numbers are never reused.	Samples
☐	14. Open MB-00012, tap Delete sample and confirm.	Refused: "A field duplicate points at this sample. Delete the duplicate first." Nothing is deleted.	Samples
☐	15. On the MB-DDH-001 hole screen, type 12x as Actual final depth and Save.	Red "Final depth must be a number above 0." Nothing is saved.	Hole
☐	16. Tap Completed on any hole with a Started date.	The calendar opens on today. Days before the Started date are greyed out and cannot be chosen.	Hole

10 Field conditions

These are the things a desk test cannot show. Do them once during the day and tick when done.

	Check	What to note
☐	No signal all day. Airplane mode was on from Part 1 to Part 9.	Did anything ask for the internet or fail because it was offline?
☐	Sunlight. Take the phone outside and read the hole screen and the sample trace at full brightness.	Can you tell a ticked step from a next step from an amber problem at a glance?
☐	Gloves or dirty hands. Enter three runs with gloves on, or with damp fingers.	Any button too small? Any mis-taps? Which ones?
☐	Interruption. Start an interval, switch to another app for a minute, come back.	Was your half-typed interval still there?
☐	Battery. Note the battery at the start and after Part 5.	Start: ____ % After Part 5: ____ %
☐	Dark and light. Switch the phone between dark and light mode once.	Is everything readable in both? Any text that vanishes?
☐	Speed. Does any screen feel slow (more than 2 seconds)?	Which screen, doing what?

Appendix: the codes used today

All of these are in the app's starter code library. You can see and edit the library from Core log → **Code library**. These are a generic hard-rock set; tell us which codes your team really uses.

Lithology

OVB	Overburden / soil
AND	Andesite
BX	Breccia
DIO	Diorite
QV	Quartz vein
GRD	Granodiorite

Alteration type

ARG	Argillic
PHY	Phyllic (sericitic)
SIL	Silicic
PROP	Propylitic

Alteration intensity

1	Weak
2	Moderate
3	Strong
4	Pervasive / intense

Mineralisation: mineral

PY	Pyrite
AU	Visible gold

Mineralisation: style

DISS	Disseminated
VNL	Veinlets
STK	Stockwork
BXH	Breccia-hosted

Weathering

FR	Fresh
SW	Slightly weathered
MW	Moderately weathered
HW	Highly weathered
CW	Completely weathered

Structure type

FLT	Fault
FRAC	Fracture zone
VEIN	Vein

Sign-off sheet

Tester _____

Date _____

Phone model and Android version _____

App version _____

Started at _____

Finished at _____

Part	What	Target (min)	Your time (min)	How smooth? 1 (painful) to 5 (effortless)	Step numbers that did not match
1	Set up project and hole	10		☐ ☐ ☐ ☐ ☐	
2	Core reception (runs and boxes)	45		☐ ☐ ☐ ☐ ☐	
3	Photograph the core	25		☐ ☐ ☐ ☐ ☐	
4	Log the core	60		☐ ☐ ☐ ☐ ☐	
5	Sample the core, with QC	75		☐ ☐ ☐ ☐ ☐	
6	Close out the hole	5		☐ ☐ ☐ ☐ ☐	
7	Trace samples	10		☐ ☐ ☐ ☐ ☐	
8	Export and check	15		☐ ☐ ☐ ☐ ☐	
9	Try to break it	20		☐ ☐ ☐ ☐ ☐	
10	Field conditions	–		☐ ☐ ☐ ☐ ☐	

What confused you or slowed you down?

What did you expect the app to do that it did not?

What would you never trust the app with in the field, and why?

Send this sheet, with screenshots of anything that did not match, to the product owner.